Heap Sort

Heap sort is a comparision-based sorting algorithm that uses a binary heap data structure to sort elements. It works by first building a max heap (for ascending order), then repeatedly swapping the largest element (root of the heap) with the last element of the array, reducing the heap size, and ~~heapifying~~ heapifying the root again.

Steps:-

1.> Build a max heap from the input data.

2.> Swap the root (maximum value) with the last item of the heap.

3.> Reduce the size of the heap by one and heapify the root.

4.> Repeat step 2 and 3 until the heap is empty.

Akash

→ // c program for heap sort;

```
#include<stdio.h>

// function to heapify a subtree rooted at index i
void heapify (int arr[], int n, int i)
{
    int largest = i; // initialize largest as root
    int left = 2*i + 1; // left child index
    int right = 2*i + 2; // right child index

// IF left child is larger than root
if (left<n && arr[left] > arr [largest])
{
    largest = left;
}

// IF right child is larger than largest so far
    if ( right < n && arr[right] > arr[largest])
{
    largest = right;
}

// IF largest is not root
if (largest != i)
{
                                    // swap (arr[i], arr[largest])
    int temp = arr [i];
    arr[i] = arr [largest];
    arr[largest] = temp;
```

Akash

```
// recursively heapify the affected sub-tree
    heapify (arr, n, largest);
  }
}


// main function to do heap sort
void heapsort (int arr[], int n)
  {
      // build max heap
      int = i;
    for (i= n/2 -1 ; i>=0; i--)
      {
          heapify (arr, n, i);
      }
    // one by one extract elements from heap
    for (i=n-1; i>0; i--)
      {                                  //swap (arr[0], arr[i])
          int temp = arr[0];
              arr[0] = arr[i];
              arr[i] = temp;

      heapify (arr, i, 0);
    }
}


int main()
  {
      int a[20], n, i;
      printf("enter size of an array");
      scanf("%d", &n);
```

```
for(i=0; i<n; i++)
  {
    printf("enter data \n");
    scanf("%d", &a[i]);
  }

heapsort (arr, n);

printf("sorted array using heap sort: \n");

for(i=0; i<n; i++)
  {
    printf("%d ", a[i]);
  }
return 0;
}
```

Akash

→ Max heap →

~~① The largest element is always at the root~~

~~② keep removing the~~

① It must be a binary tree (each node has at most 2 children).

② Complete tree → All levels are completely filled except possibly the last, which is filled from left to right

③ Max-heap property
for every node i

- value(i) ≥ value(left child)
- value(i) ≥ value(right child)

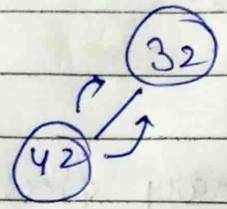
Akash

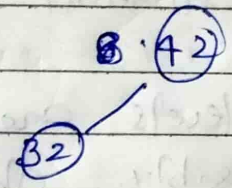
Q. Describe the insertion in max heap tree with example from the following list of numbers: 32, 42, 67, 23, 44, 49, 74.

=>

Step 1 → 32 add, 32 is root

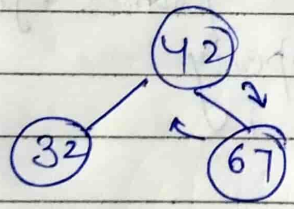
2 → add 42; 32 < 42, swap

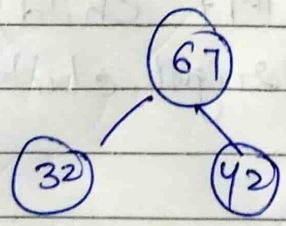




add 67;

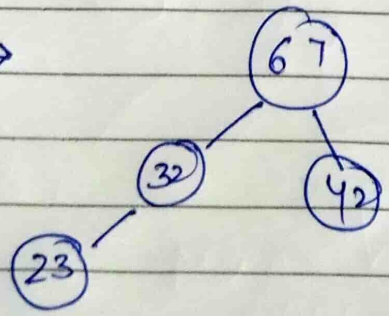
3 → 42 < 67, swap





add 23;

4 → 32 > 23, OK



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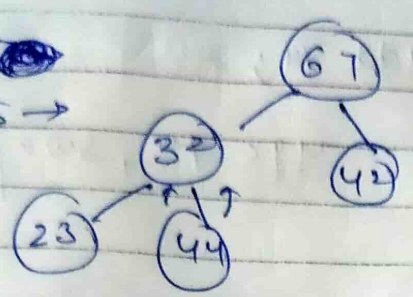
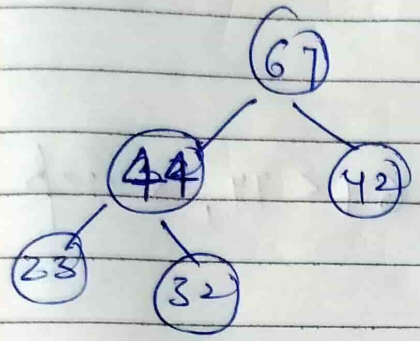
5 →

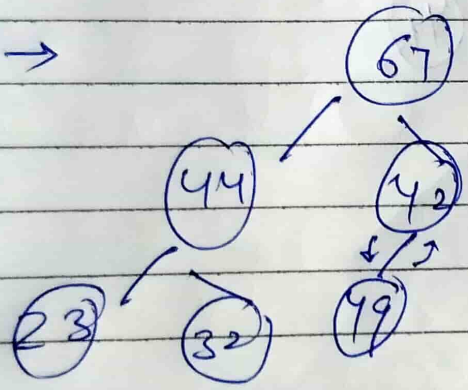


add 44;

32 < 44, swap

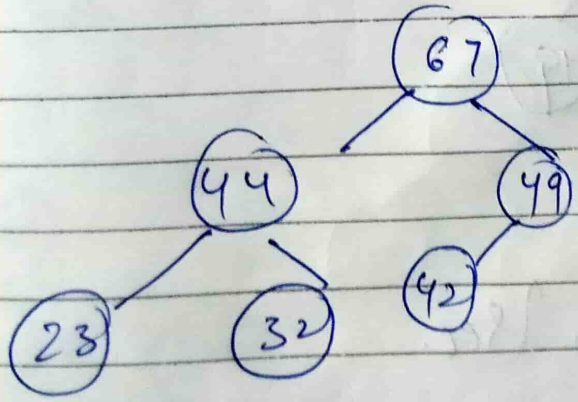


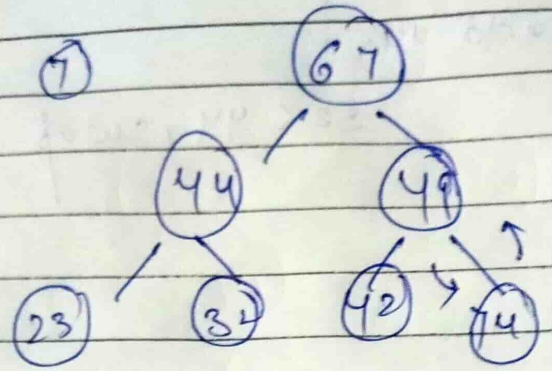
6 →



add 49;

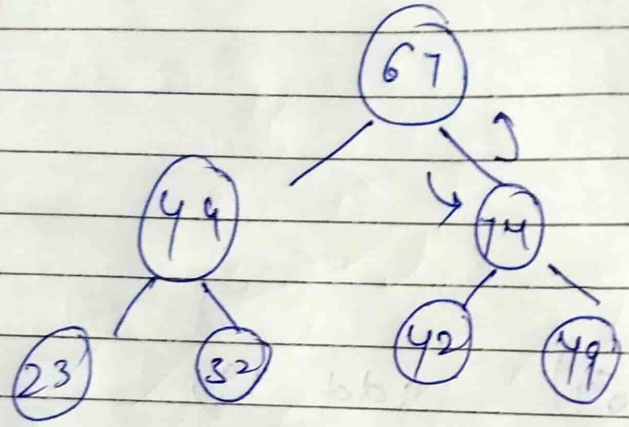
42 < 49, swap



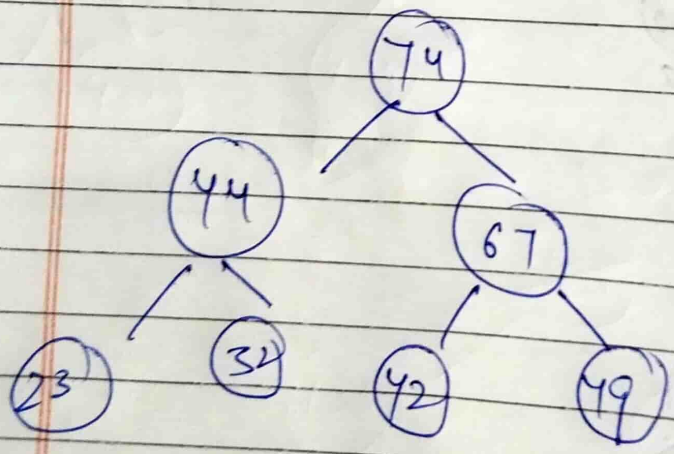


add 74;

49 < 74, swap



67 < 74, swap



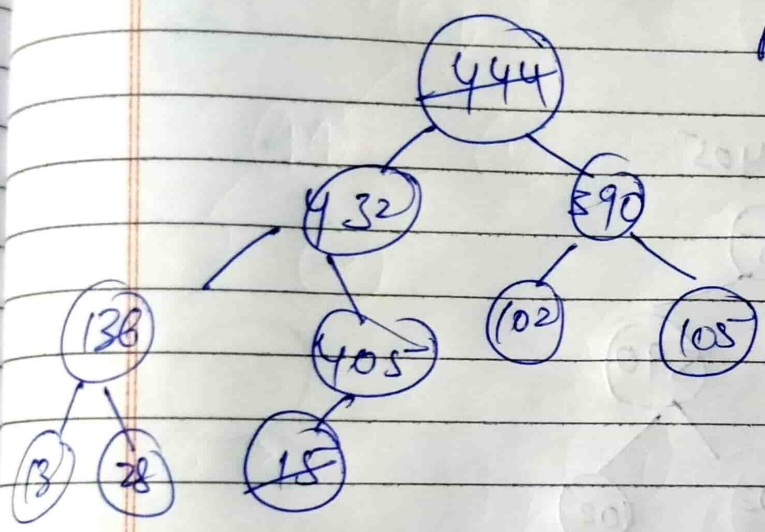
Max-heap Tree

Akash

→ Eg; for heap sort;

13, 102, 405, 136, 15, 105, 390, 432, 28, 444.

⇒ Max-heap tree :-

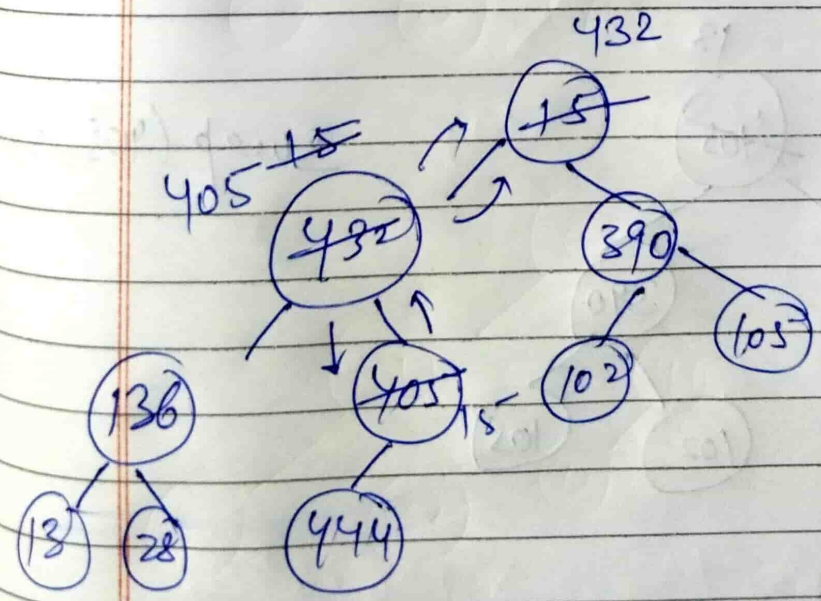


Now, heapifying

Swap (a[last], a[0])

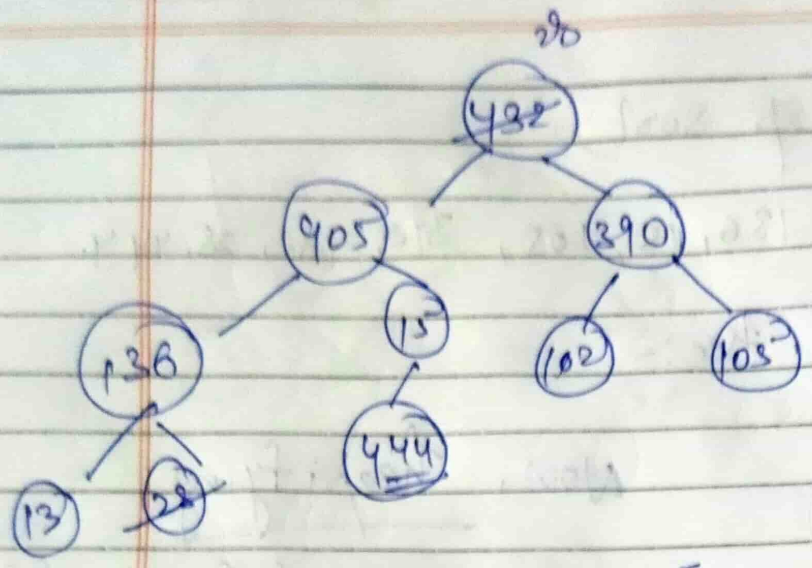
Swap (444, 15)

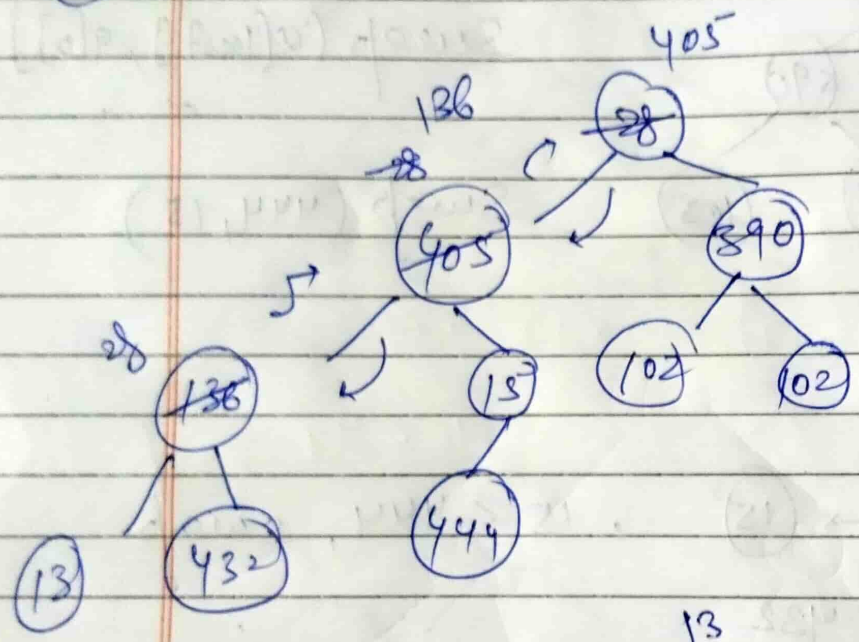
last one → 15 , 15 < 444, swap



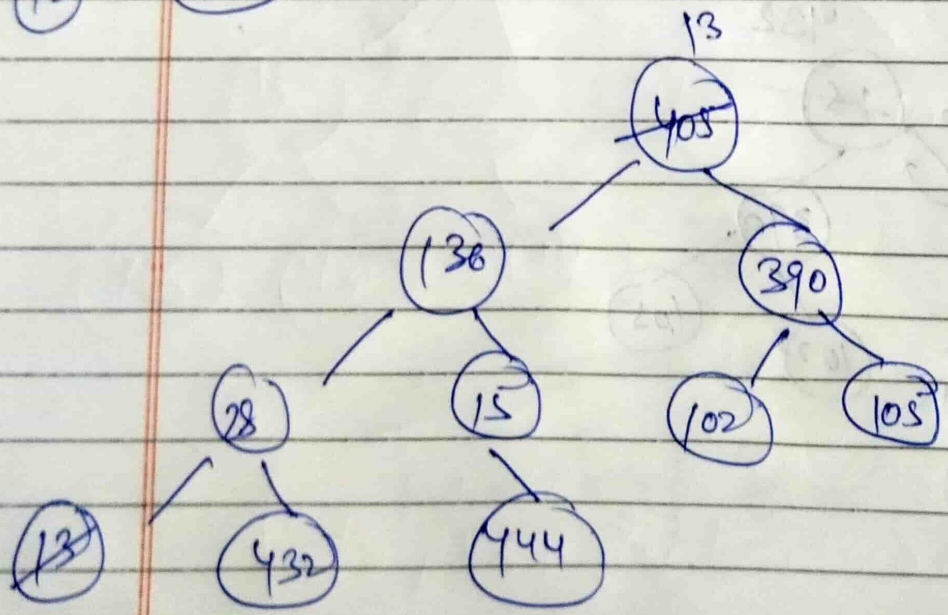
Akash

swap (28, 432)

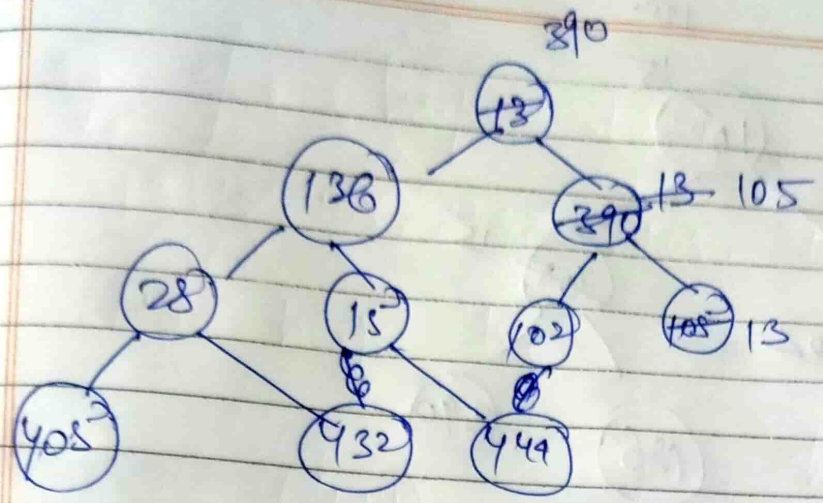




swap (405, 13)



Akash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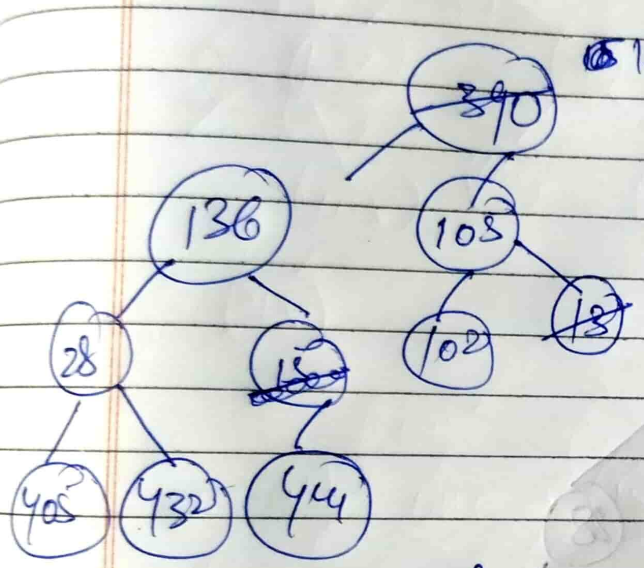
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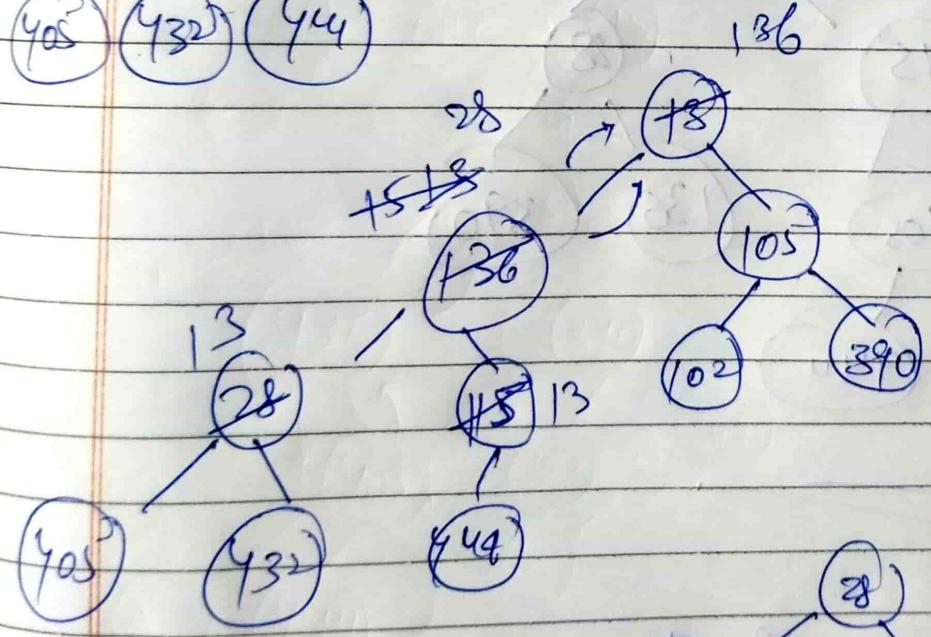
444

swap (13, 390)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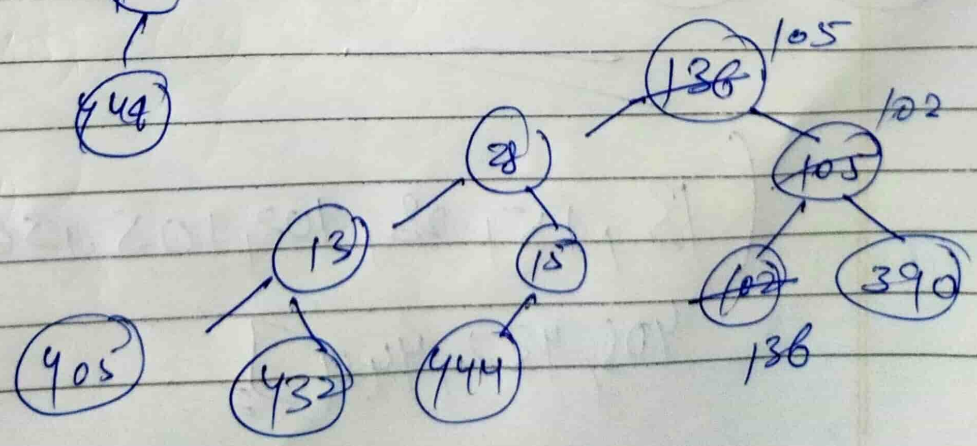
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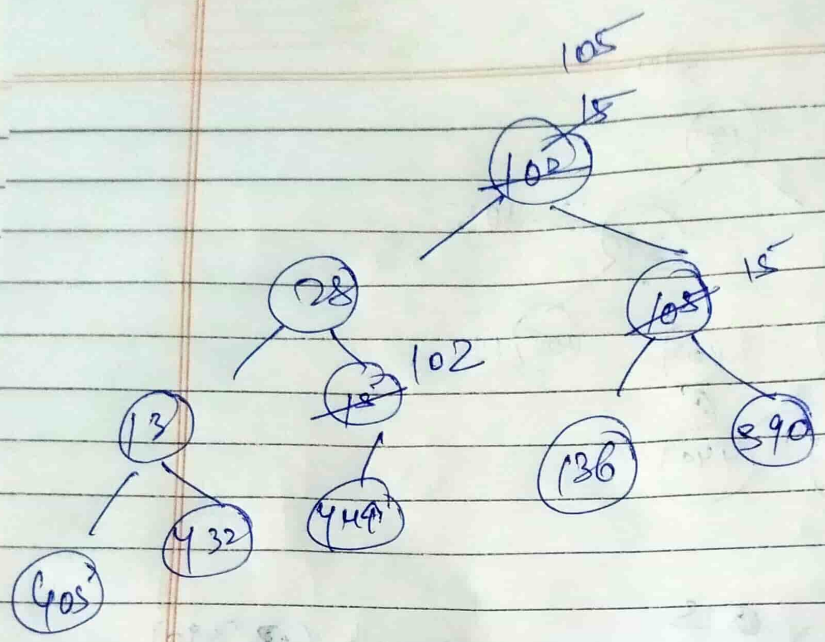
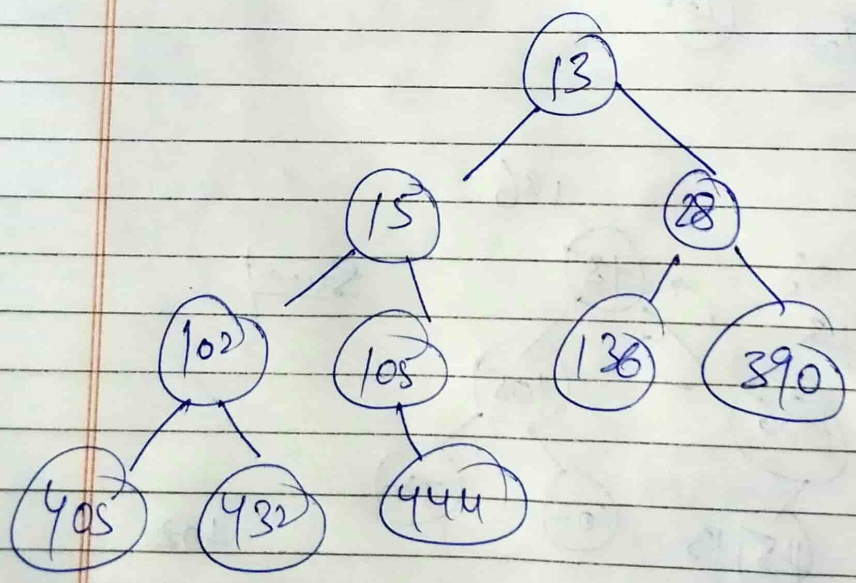


Similar



[13, 15, 28, 102, 105, 136, 390, 405, 432, 444]

Time Complexity

Case	Time Complexity
Best	$O(n \log n)$
Average	$O(n \log n)$
Worst	$O(n \log n)$

Akash

Akash